Do Prediction Markets Know What the Fed Will Do? Kalshi and Polymarket versus Futures, the Dollar and Treasury Bills, 2023–2026

Speed, tradability, cross-venue prices and forecast accuracy of Fed-decision markets

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ABSTRACT

Kalshi and Polymarket let anyone bet on the next Federal Reserve decision, and their prices read as probabilities. This paper asks how good those probabilities are and how fast they absorb macroeconomic news, using every Fed-decision market on both venues from May 2023 to October 2026 (31 meetings, 28 decided), every trade around 81 CPI and payrolls releases, 1-second S&P 500 futures and USD/JPY data, and daily Treasury yields. Four data problems are found and fixed before any result is estimated, and both venues are shown to have paid out the true decision at all 28 meetings. After a release, S&P 500 futures and USD/JPY complete half of their five-minute reaction in 3–4 seconds; Kalshi needs about 41 seconds and Polymarket about 30 (September 2025 to October 2026), down from 95–110 seconds in 2025. The lag is tradable for a fast participant: reading USD/JPY's first five seconds and trading the next meeting's favourite outcome earns 3–4 cents per contract after spread and fees, 5–7 cents when the decision is a close call, but nothing distinguishable from zero after 30 seconds, and only a few thousand dollars trade at those prices. Neither venue leads the other, and their prices for the same outcome leave a locked-in profit after costs in about a fifth of hours, by half a cent, a share that fell from 27% in 2024 to 13% in 2026. The forecasts are accurate (the favourite outcome happened at 28 of 28 meetings one day ahead and 27 of 28 one week ahead) and increasingly tied to the 6-month Treasury bill.

Keywords: prediction markets, monetary policy expectations, price discovery, macroeconomic announcements, market efficiency, limits to arbitrage. **JEL:** G14, E52, E44, D84.

1. Introduction

Markets price the next Federal Reserve decision continuously. Professional investors read it from fed funds and SOFR futures and from short-term Treasury yields (Kuttner, 2001). Since 2023 two prediction markets have offered something simpler: a contract for each possible decision (cut, hold, hike) that pays one dollar if it happens. Kalshi is a US-regulated exchange; Polymarket runs on a blockchain. By 2026 Polymarket alone traded $140–660 million on each meeting, and their probabilities are quoted in the financial press.

Wolfers and Zitzewitz (2004) argue that prediction markets aggregate information efficiently. Two features make the Fed markets a good test. First, the information arrives at known times: the CPI and payrolls reports are released at 08:30 New York, so the speed of the reaction can be compared, second by second, with markets run by professional traders. Second, the outcome is unambiguous, so forecast accuracy can be scored exactly.

This paper asks four questions. (i) How fast do the two venues reprice the next Fed decisions after CPI and payrolls, compared with S&P 500 futures and USD/JPY? (ii) Is any lag large enough to profit from after spreads and fees? (iii) Does one venue lead the other, and do their prices for the same outcome differ by more than the cost of trading the gap? (iv) How accurate are their forecasts, and do they move with the bond market?

Main findings. (1) The data can be trusted once four problems are fixed; both venues paid out the true decision at all 28 meetings and agree hour by hour (correlation 0.994). (2) Prediction markets take 30–40 seconds to do half their reaction to CPI and payrolls, against 3–4 seconds for futures and the dollar, down from about 100 seconds in 2025. (3) The lag is worth 3–4 cents per contract after costs to someone acting within seconds, and nothing after 30 seconds. (4) Neither venue leads; their price gaps exceed costs in a shrinking fifth of hours. (5) Forecasts are accurate and increasingly tied to Treasury bills.

The contributions are three. **A validated data set:** both venues' public data contain traps (prices in the wrong units, late time stamps, empty order books) that change results if missed; each is documented with the check that found it. **A speed comparison at second resolution** between retail-facing prediction markets and professional markets, using exact trade times. **An economic test** that turns the lag and the cross-venue gaps into cents per contract after real fees.

Section 2 describes the markets and data, Section 3 the validation, Sections 4–8 the results, Section 9 robustness and limitations, and Section 10 concludes.

2. Markets and data

Contracts. For each Federal Open Market Committee (FOMC) meeting, Kalshi lists five yes/no contracts (cut by more than 25 bp, cut 25, hold, hike 25, hike by more than 25) and Polymarket four to six similar ones. A YES contract pays $1 if its outcome happens; its price, between 0 and 1, reads as the market's probability. Kalshi charges takers a fee of $0.07 \times p(1 - p)$ dollars per contract (1.75 cents at $p = 0.5$); Polymarket charged no fees on these markets until 2026 and then a rate of $0.05 \times p(1 - p)$. Kalshi's contracts for a meeting stop trading at 13:55–13:59 New York, just before the 14:00 decision, so the reaction to a decision appears in the next meeting's contracts.

One number per venue. For a meeting with outcome probabilities p_k and rate changes Δ_k (in basis points), the expected rate change is

$$E_t = \Sigma_k \, p_{k,t} \, \Delta_k \, / \, \Sigma_k \, p_{k,t} \tag{1}$$

Dividing by the sum removes the small overround (prices add up to slightly more than 1). An outcome nobody quotes counts as zero when the quoted outcomes already add up to 97% or more. Kalshi prices are quote mid-points (the last trade if the quote is wider than 10 cents); Polymarket prices are its price history. Around releases, E is summed over the next two meetings that both venues list.

Table 1. Sample.

Data	Source	Coverage
Fed-decision markets, hourly prices	Kalshi, Polymarket public APIs	31 meetings, May 2023 – Jan 2027
1-minute prices and every trade around releases	Kalshi, Polymarket	109 windows (40 CPI, 41 payrolls, 28 FOMC)
trades in those windows		99,320 Kalshi, 227,448 Polymarket
S&P 500 futures, USD/JPY, 1-minute	London Strategic Edge	May 2023 – Oct 2026
S&P 500 futures, USD/JPY, 1-second	London Strategic Edge	41 releases since Jan 2025 (futures: 22, from Sep 2025)
Macro calendar (CPI, payrolls, FOMC)	London Strategic Edge	Official release times, 2023–2026
Daily rates: fed funds, T-bills, 1-, 2-, 10-year	FRED	793 business days

Polymarket traded $3.6 billion on the main event of these 31 meetings, most of it since 2025. Release windows run from 3 hours before to 6 hours after each release; second-level data from 5 minutes before to 10 minutes after.

3. Validation: four problems and four checks

Every result in this paper rests on the data being right, so the data were tested first. Four problems were found, each by a check that failed (Table 2).

Table 2. Data problems found and fixed.

Problem	How it was found	Fix
Kalshi's historical endpoint sends prices in dollars without the "_dollars" label; read as cents, prices before Aug 2026 were 100× too small	A meeting's outcome prices added up to 0.01, not 1	Read as dollars; unit tests for all three formats
Polymarket stamps its 1-minute prices 2–11 s into each minute	On its own stamps its biggest move after CPI and payrolls never fell in the release minute (0%, vs 77% for futures)	Each minute price read as the start of its minute; second-level work uses trade times
2023 Kalshi tickers contain ">" (e.g. H>25)	The download failed on Windows	Safe file names and URL escaping; unit test
Many 2023–24 Polymarket books were empty (price fixed at 0.50)	Cross-venue "gaps" of 20–40 cents	Gap tests use liquid meetings only (Section 7)

Four checks then test the whole data set. **Settlement:** at all 28 decided meetings both venues paid out the same outcome, and it matches the change in the effective fed funds rate (19 holds, 5 cuts of 25 bp, one cut of 50, three hikes of 25). **Coherence:** each venue's outcome prices add up to about one (median 1.015 on Kalshi, 1.003 on Polymarket). **Agreement:** over 33,275 common hours the two venues' expected rate change correlates 0.994, with a median gap of 0.38 bp (Figure 1). **Clocks:** around CPI and payrolls, every market's biggest one-minute move falls in the release minute far more often than in any other (S&P futures 77% of releases, USD/JPY 73%; the prediction markets' moves are spread over the following minutes, the subject of Section 4).

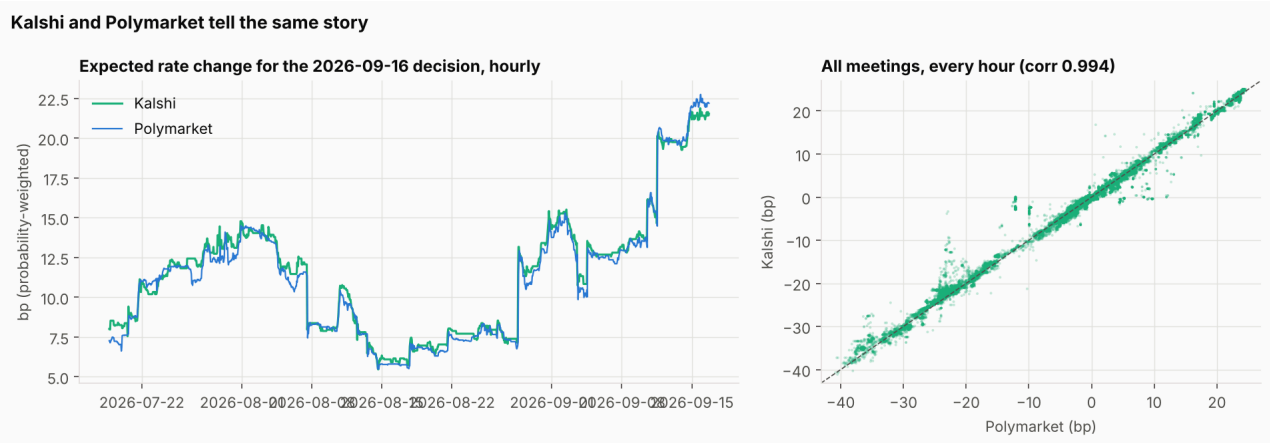


Figure 1. Left: expected rate change for the 16 September 2026 decision, hourly, on both venues. Right: every common hour of every meeting.

4. How fast is the news priced? Minute by minute

For each release i and market X (each venue's expected rate change, or the log price of S&P 500 futures or USD/JPY, in bp), the reaction $R_i = X_i(t_0 + H) - X_i(t_0)$ is measured over $H = 30$ minutes. The share of it done after h minutes, pooled over releases, is

$$S(h) = \Sigma_i \operatorname{sign}(R_i)\,[X_i(t_0 + h) - X_i(t_0)] \;/\; \Sigma_i\,|R_i| \qquad (2)$$

Pooling this way weights large reactions more, so a release that barely moved a market adds little noise. Confidence intervals come from 2,000 bootstrap draws of releases, and only the 66 releases where all four markets have data are used, so the curves compare the same events.

Table 3. Share of the 30-minute reaction done after *h* minutes (95% CI), 66 CPI and payrolls releases, May 2023 – Oct 2026.

Market	1 minute	2 min	5 min	10 min	15 min
Kalshi	0.34 [0.22, 0.48]	0.52	0.70	0.84	0.89
Polymarket	0.47 [0.33, 0.62]	0.61	0.69	0.80	0.87
S&P 500 futures	0.64 [0.47, 0.81]	0.68	0.76	0.78	0.81
USD/JPY	0.56 [0.40, 0.74]	0.56	0.70	0.72	0.87

Kalshi minus USD/JPY in the first minute: −0.23 [−0.44, +0.02]; Polymarket minus USD/JPY: −0.10 [−0.31, +0.15]. First-minute shares, 2023–24 (n = 26): Kalshi 0.28, Polymarket 0.28, futures 0.60, USD/JPY 0.53; 2025–26 (n = 40): 0.36, 0.53, 0.67, 0.60.

The prediction markets do a third to a half of their reaction in the release minute, against about 60% for futures and the dollar, and catch up within five minutes. With one-minute prices the difference is borderline. It is also bounded by a measurement problem: Polymarket stamps its minute prices a few seconds late, and on its own stamps its first-minute share would be 0.03. Minute data cannot settle the first minute; trades can.

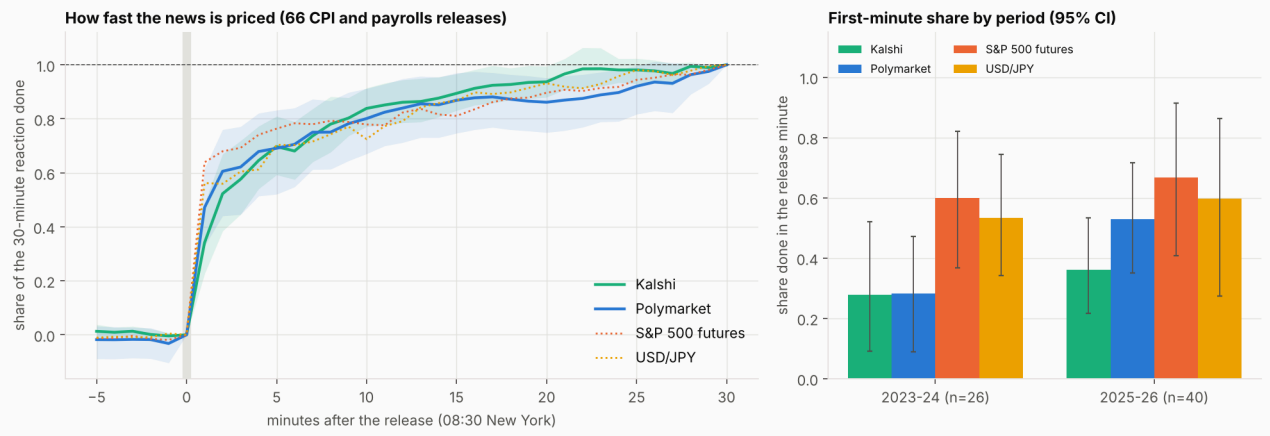


Figure 2. Left: share of the 30-minute reaction done after each minute (bands: 95% CI for the prediction markets). Right: share done in the release minute, 2023–24 and 2025–26.

5. The first minutes, second by second

Both venues record every trade with an exact time (Kalshi to the microsecond). Each outcome's probability is set to its latest trade price (a Polymarket trade in the NO token at price q means YES = 1 − q), starting from its minute price five minutes before the release. S&P 500 futures and USD/JPY come from 1-second bars. The reaction is measured over five minutes and the half-life is the number of seconds until half of it is done.

Table 4. Seconds until half of the 5-minute reaction is done (95% CI).

Sample	USD/JPY	S&P 500 futures	Polymarket	Kalshi
All four markets, Sep 2025 – Oct 2026 (22)	3 [2, 9]	4 [2, 10]	30 [21, 49]	41 [19, 70]
2025 (21)	3 [2, 4]	–	110 [74, 128]	93 [47, 187]
2026 (20)	2 [2, 8]	–	30 [21, 52]	40 [19, 65]

Share done after 10 seconds (all four markets): futures 0.71, USD/JPY 0.72, Polymarket 0.19, Kalshi 0.16. First trade after 08:30:00 in the next meeting's markets (median): Kalshi 13.9 s in 2025 and 11.7 s in 2026; Polymarket 44 s and 10 s.

The gap is large and clear: futures and the dollar finish most of their reaction within ten seconds, the prediction markets within one to two minutes. It narrowed sharply in one year as trading grew; in 2026 Polymarket averages about 50 trades in the minute after a release, against 5 in 2025 (Figure 3, right). A trade-based measure is slightly pessimistic, because quotes can move before anyone trades, but the order of magnitude is not in doubt.

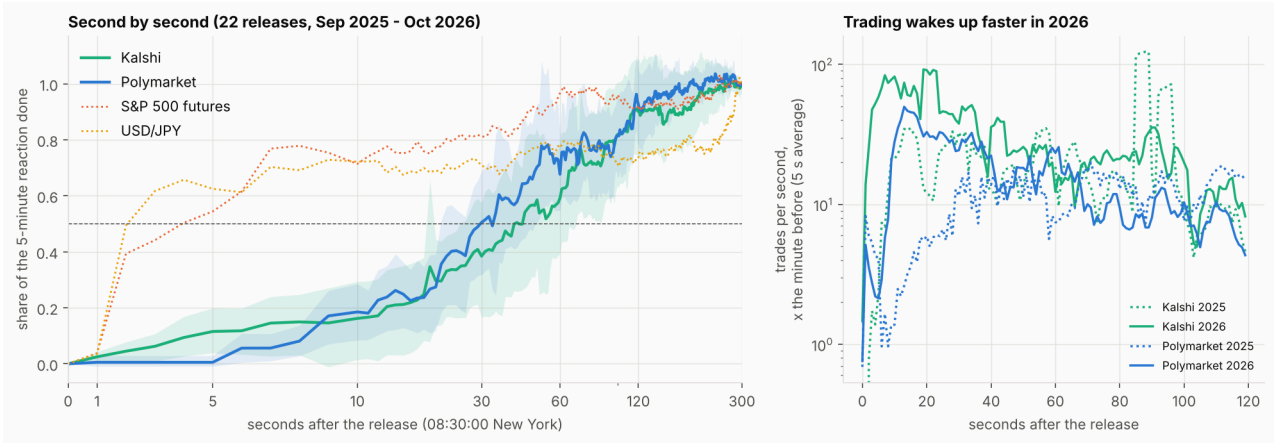


Figure 3. Left: share of the 5-minute reaction done after each second (logarithmic time axis; bands: 95% CI). Right: trades per second after the release, relative to the minute before.

6. Can the lag be traded?

A lag matters economically only if someone can earn more than the cost of exploiting it. The test mimics a fast trader. **Signal:** USD/JPY's move in the first five seconds after 08:30:00 (a stronger dollar means news that raises expected rates). The rule acts only if the move exceeds three times USD/JPY's normal five-second move in the preceding five minutes; fixed in advance, it never binds (the smallest move was six times normal). **Contract:** the favourite outcome of the next meeting, the most liquid one, bought (or sold) if the news makes it more (less) likely. **Execution:** the decision is taken at +6 seconds and filled at the first real trade on the same side after that, so the trader pays the spread; the position is closed five minutes later at the first trade on the other side. Both venues' taker fees are paid on entry and exit.

Table 5. Net profit per contract after spread and fees, cents (95% bootstrap CI), 41 releases since January 2025.

Variant	Kalshi	Polymarket
Act at 6 s, exit at 5 min (main)	+3.2 [0.5, 5.9]; 26 trades, 62% won	+4.2 [1.7, 6.7]; 29 trades, 79% won
favourite below 85% (a close call)	+5.5 [2.0, 8.8]	+7.3 [3.8, 10.7]
favourite 85% or more	−1.2 [−2.7, 0.2]	−0.2 [−1.5, 0.8]
Act at 15 s	+1.6 [−0.3, 3.5]	+2.2 [0.3, 4.1]
Act at 30 s	+0.7 [−1.2, 2.6]	+1.1 [−0.2, 2.6]
Act at 60 s	+0.3 [−1.3, 2.1]	+0.7 [−0.4, 1.9]
Exit at 15 min	+3.8 [0.3, 7.6]	+5.5 [2.1, 9.3]
One cent worse on entry and exit	+1.2 [−1.5, 3.9]	+2.2 [−0.3, 4.7]
Opposite trade (sanity check)	−7.0 [−9.5, −4.6]	−5.2 [−8.1, −2.6]

Average fees: Kalshi 2.0 cents per round trip, Polymarket 0.6 (zero before 2026). Volume traded on our side within 30 s of entry at no more than 2 cents worse: a median of about $6,500 (Kalshi) and $13,500 (Polymarket) per release.

Three results stand out. The edge exists only for someone acting within about 15 seconds; after 30 seconds it cannot be told apart from zero. It exists only when the next decision is a close call: a contract already priced at 97% can rise by three cents at most. And it is small in money: a few thousand dollars trade at those prices, so the profit is a few hundred dollars per release at best. This is the Grossman and Stiglitz (1980) equilibrium seen in the companion study of index futures: the lag survives up to the cost and capacity of exploiting it, and the fastest participants compete it away (Budish, Cramton and Shim, 2015). The 2025 to 2026 comparison points the same way: fills came after a median 33 seconds in 2025 and 10 seconds in 2026, while USD/JPY's five-second moves were if anything smaller (17.6 against 26.2 bp).

Design note. A first version traded the most uncertain meeting, often a later one whose contracts barely traded at 08:30, and most orders could not be filled. The next meeting's favourite was chosen for liquidity before profits

were examined, and every variant tried is reported in Table 5.

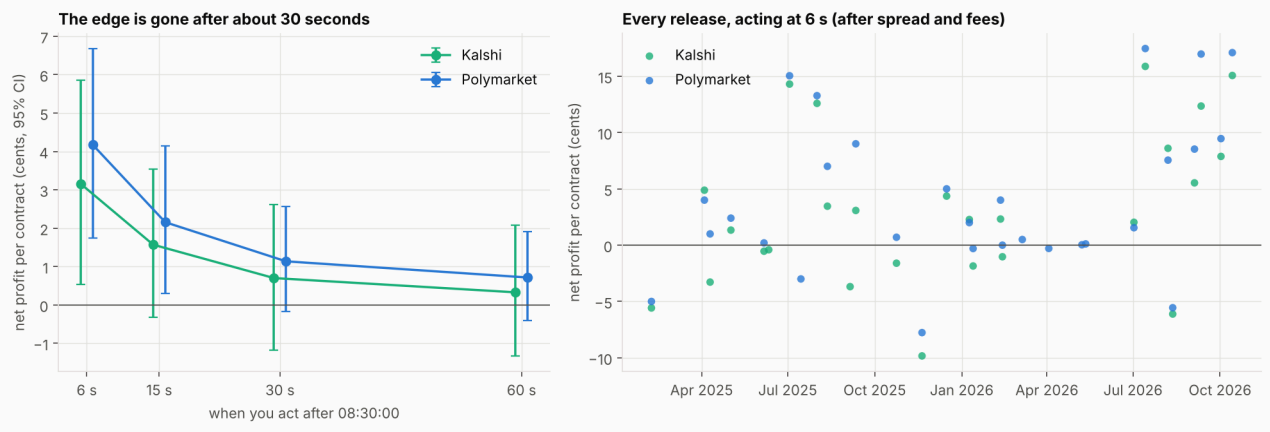


Figure 4. Left: net profit per contract by how soon after 08:30:00 the trader acts (95% CI). Right: every release, acting at 6 seconds.

7. Kalshi versus Polymarket

Who moves first? On 2-, 5- and 10-second changes in the first five minutes after each release the lead-lag asymmetry is

$$A = \mathrm{corr}(\Delta K_t, \Delta P_{t+1}) - \mathrm{corr}(\Delta P_t, \Delta K_{t+1}) \tag{3}$$

where K and P are the two venues' expected rate changes ($A > 0$: Kalshi first). A venue that trades less often keeps an old price for longer and looks like a follower even when it is not (the staleness trap; Epps, 1979). Polymarket trades about twice as often as Kalshi at 08:30, so only intervals in which both venues traded are used. The asymmetry is +0.10 [−0.02, +0.16] at 5 seconds, changes sign at 2 seconds (−0.08 [−0.30, +0.08]) and is close to zero at 10 seconds (+0.05 [−0.10, +0.15]); without the filter the results are similar. On the 14 releases where both venues moved by at least 1 bp, the median half-life is 54 seconds on Kalshi and 62 on Polymarket, the paired difference −1 second [−13, +10], and each venue was faster in about half of them (7 against 6). Their five-minute reactions correlate 0.94. Neither venue leads.

Do prices for the same outcome differ by more than costs? Buying YES on the cheaper venue and NO on the dearer one pays exactly $1 at the decision, since the venues settled all 28 meetings identically. Buying YES on Kalshi at the ask a and NO on Polymarket at $1 - p + s$ locks in

$$\pi = 1 - a - (1 - p + s) - f_{\mathrm{K}}(a) - f_{\mathrm{P}}(p - s) \tag{4}$$

and symmetrically in the other direction, where f are the taker fees and s an assumed Polymarket half-spread (0.5 cents; its price history has no quotes). The sample is hourly over the last 60 days before each decision, for outcomes priced 3–97%, restricted to liquid meetings (at least $10 million traded on Polymarket, from September 2024) and hours when Kalshi's spread is at most 5 cents: 35,568 outcome-hours over 18 meetings.

Table 6. Cross-venue gaps, liquid meetings.

Period	Average price gap	Hours with a locked-in profit	Median profit	90th percentile
2024	1.84 c	27.3%	0.97 c	2.73 c
2025	1.28 c	22.5%	0.52 c	1.76 c
2026	1.18 c	12.8%	0.43 c	1.43 c
All	1.32 c	19.5%	0.54 c	1.92 c

With a 1-cent Polymarket half-spread the share of hours falls to 10.5% (2026: 6.1%). Including the illiquid 2023–24 meetings raises the 2024 average gap to 3.6 cents and its 90th percentile to 21 cents, mostly from empty order books.

The gaps are small and closing. They persist partly for reasons outside the model, the limits to arbitrage of Shleifer and Vishny (1997): few traders can use both venues (one is a US-regulated exchange, the other runs on

crypto), the money is tied up until the decision, and the depth at those prices is unknown.

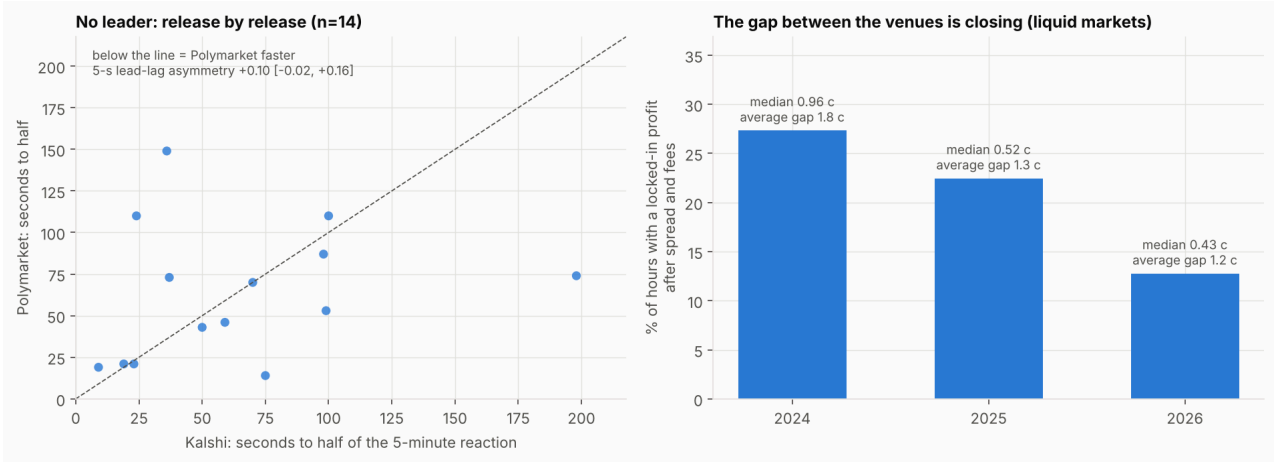


Figure 5. Left: half-life of each venue's reaction, release by release (below the line: Polymarket faster). Right: share of liquid hours with a locked-in cross-venue profit after costs, by year.

8. Accuracy and the bond market

Accuracy. For each decided meeting the probabilities each venue gave 1 day, 1 week, 1 month and 2 months before the decision are rescaled to add up to one and scored with the Brier (1950) score, $B = \Sigma_k\ (p_k - 1[k \text{ happened}])^2$, where 0 is perfect.

Table 7. Forecast accuracy, 28 decided meetings.

Before the decision	Favourite happened (Kalshi / Polymarket)	Median probability of what happened	Brier score
1 day	100% / 100%	0.98 / 0.98	0.021 / 0.021
1 week	96% / 96%	0.96 / 0.96	0.090 / 0.087
1 month	93% / 92%	0.84 / 0.81	0.150 / 0.168
2 months	88% / 82% (8 / 17 meetings)	0.75 / 0.68	0.223 / 0.267

Benchmarks without market information: "the Fed holds" is right at 68% of meetings, "the Fed repeats its last decision" at 67%. The one miss a week ahead: on 18 September 2024 both venues gave 90–93% to a 25 bp cut; the Fed cut by 50 (priced at about 53% the day before).

The two venues are equally accurate. Calibration can only be checked roughly with 28 meetings: a month ahead, outcomes priced at 20–40% happened 13% of the time (n = 15, 95% interval 4–38%) and outcomes priced at 60–80% happened 85% of the time. That pattern is the favourite–longshot bias of betting markets (Thaler and Ziemba, 1988), but the sample cannot confirm it.

The bond market. Each business day at 16:00 New York the day-on-day change in the expected rate change over the next two decisions is compared with the day's change in Treasury yields (793 days). It correlates 0.56 (Kalshi) and 0.46 (Polymarket) with the 6-month bill, with a slope of about 0.4 bp per bp (Newey–West t = 8.7 and 7.7), and less with longer maturities (2-year: 0.41 and 0.32). The link has tightened every year: from 0.37 and 0.23 in 2023 to 0.66 and 0.61 in 2025, and 0.64 and 0.58 in 2026. From one day to the next neither moves first in any meaningful way (all lead and lag correlations are 0.11 or less).

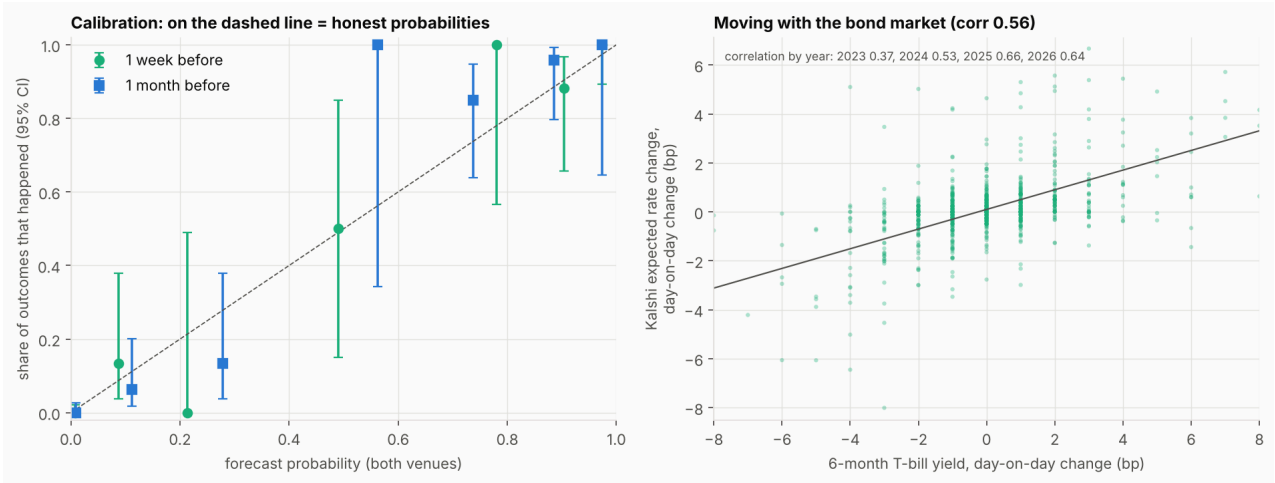


Figure 6. Left: calibration, both venues pooled (95% Wilson intervals); on the dashed line probabilities are honest. Right: daily changes of Kalshi's expected rate change against the 6-month T-bill.

9. Robustness and limitations

- **Measurement choices.** The speed results hold with a 60-minute reaction window and in the half of releases with the biggest news (Section 4); the trading results with later entries, later exits and worse fills, all reported in Table 5; the gap results with a wider assumed Polymarket spread.
- **Small samples.** 28 decided meetings, 81 releases, 41 with second-level data and 22 with second-level futures. Every estimate carries a confidence interval and several are borderline.
- **Trades, not order books.** Speeds and fills are measured from trades. Quotes can move before trades and the depth behind each price is unknown, so the trade-based speed is slightly pessimistic and the trading capacity an estimate.
- **No rate futures.** The data vendor has no fed funds or SOFR futures, the professional benchmark for Fed expectations; the comparison uses futures and the dollar intraday and Treasury bills daily.
- **Access.** Kalshi and Polymarket serve different users. The gap test treats this as a reason gaps persist, not as a cost it can price.

10. Conclusion

Fed-decision prediction markets are good forecasters and slow traders. Their probabilities picked the right decision at every meeting one day ahead and at all but one a week ahead, and they move more and more with the Treasury bill market. But after CPI and payrolls they take 30–40 seconds to do what futures and the dollar do in three or four. That lag can be traded for a few cents per contract, only within seconds, only when the decision is a close call and only in small size, which is why it survives. Kalshi and Polymarket are equally fast and their price gaps are small and shrinking.

The trend matters more than any single number. Between 2025 and 2026, as trading grew, the half-life of the reaction fell by about two thirds, the first Polymarket trade after a release came four times sooner, and the share of hours with a cross-venue profit halved. These markets are converging on the efficiency of the professional markets they shadow. Natural extensions are order-book data, to measure speed from quotes rather than trades, and fed funds futures, the benchmark these markets are ultimately competing with.

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Appendix: reproducibility

The full pipeline is public at github.com/unaifmm/prediction-markets-fed. Numbered Python scripts run in order, from the data download (`01_download_data.py`) to the accuracy and bond-market analysis (`07_accuracy_bonds.py`), and `python run_all.py --download` reproduces every figure and table in this paper. Twenty unit tests, using fake versions of the venues' APIs, check the building blocks against cases with a known answer and run automatically on GitHub at every change. Kalshi, Polymarket and FRED data are public and need no account; London Strategic Edge data need a subscription. Raw data are not included. The Python code was written with the help of AI coding assistants; the research questions, design and interpretation are the author's, and every result was checked against the data before it was used. This paper is for research and educational purposes only and is not investment advice.